Nullforming with Distributed Arrays and Applications to Satellite Anti-Jamming

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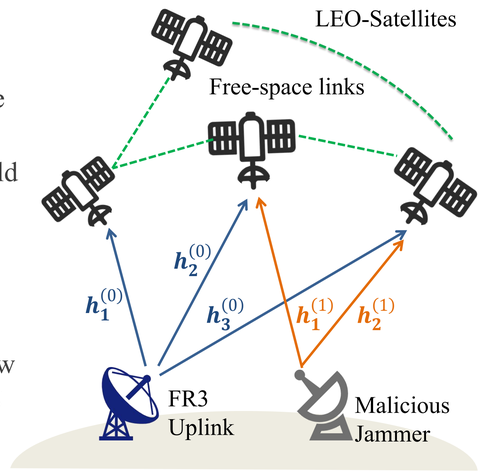
Motivation

❑ **Problem:**

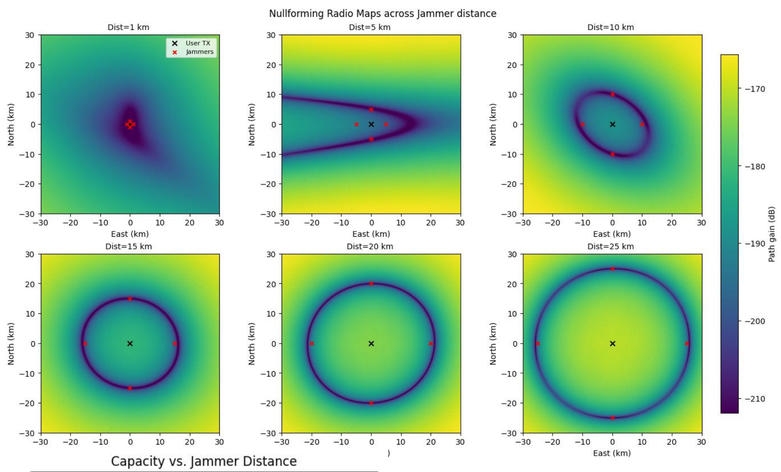
- Single satellite is vernable under jammer attack.
- Distirbuted satellites would break synchronization

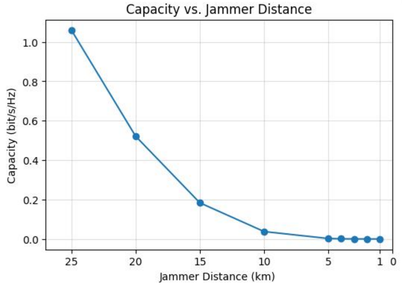
❑ **Challenges:**

- Estimation the channel $\boldsymbol{h}$ and delays $\boldsymbol{\tau}$
- Given gain and delay, How do we perform the nulling



When Jammer too close, single satellite can't null !





❑ Satellite nulls distant jammers [2]:

$$\widehat{\boldsymbol{w}} = \arg\max_{\|\boldsymbol{w}\|=1} \left[\left| \boldsymbol{w}^* \boldsymbol{h}_m^{(0)} \right|^2 - \lambda \sum_{i=1}^{K} \left| \boldsymbol{w}^* \boldsymbol{h}_m^{(1)} \right|^2 \right]$$

❑ For close jammer, also suppress the desired signal.

❑ Under 5 km, ineffective.

1. J. Hoydis, S. Cammerer, F. Ait Aoudia, M. Nimier-David, L. Maggi, G. Marcus, A. Vem, and A. Keller, "Sionna," 2022, https://nvlabs.github.io/sionna/.
2. S. Jia, M. Ying, M. Mezzavilla, D. Calin, T. S. Rappaport, and S. Rangan,"Joint Detection, Channel Estimation and Interference Nulling for Terrestrial-Satellite Downlink Co-Existence in the Upper Mid-Band,"arXiv preprint arXiv:2510.08824, Oct. 2025.

Distributed Arrays Min - Max Game

❑ **System Model:** Consider a desired and a jammer with array antennas.

The received signal at satellite $\boldsymbol{m}$ is

$$r_m[k] = \boldsymbol{h}_m^{(0)} \boldsymbol{x}_0[k] + \boldsymbol{h}_m^{(1)} \boldsymbol{x}_1[k] + d_m[k].$$

❑ Both transmitters apply beamforming with power constraints:

$$\boldsymbol{x}_i[k] = \boldsymbol{w}_i s_i[k],\ \boldsymbol{Q}_i = \mathbb{E}\left[\boldsymbol{x}_i[k]\boldsymbol{x}_i^{\mathrm{H}}[k]\right],\ \mathrm{Tr}(\boldsymbol{Q}_i) = \mathcal{E}_i.$$

❑ Stacking all receivers:

$$\boldsymbol{r}[k] = \boldsymbol{H}_0 \boldsymbol{w}_0 s_0[k] + \boldsymbol{H}_1 \boldsymbol{x}_1[k] + \boldsymbol{d}[k],$$
$$\boldsymbol{P}(Q_1) = H_1 Q_1 H_1^{\mathrm{H}} + N_0 \boldsymbol{I}.$$

❑ The achievable rate is:

$$J(\boldsymbol{Q}_0, \boldsymbol{Q}_1) = \log_2 \det\left(\boldsymbol{I} + \boldsymbol{H}_0 \boldsymbol{Q}_0 \boldsymbol{H}_0^{\mathrm{H}} \boldsymbol{P}^{-1}\right)$$

❑ Leading to the minimax problem:

$$\begin{cases} \widehat{\boldsymbol{Q}}_1 = \arg\min_{\boldsymbol{Q}_1} J(\widehat{\boldsymbol{Q}}_0, \boldsymbol{Q}_1), \\ \widehat{\boldsymbol{Q}}_0 = \arg\max_{\boldsymbol{Q}_0} J(\boldsymbol{Q}_0, \widehat{\boldsymbol{Q}}_1), \end{cases}$$

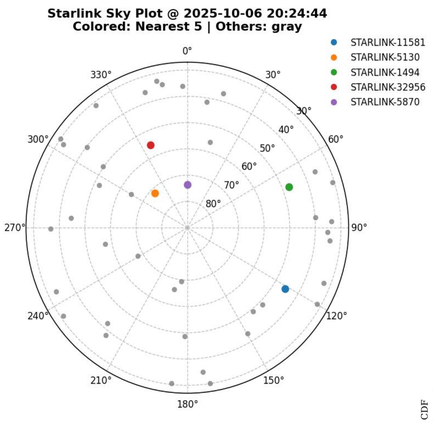
Simulation Settings

Algorithm 1: Best-Response + Mirror Descent for NTN Min–Max Game

Input: H_0, H_1, N_0, P_0, P_1, step η, tolerance ε
Initialize: $Q_0^{(0)} = \frac{P_0}{N_0} I$, $Q_1^{(0)} = \frac{P_1}{N_1} I$

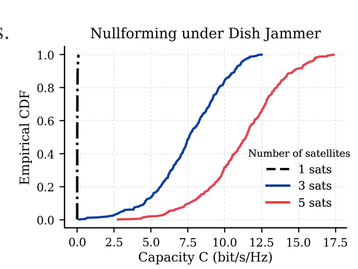
for $t = 0, 1, \ldots$ **do**
- // (A) Best response of Q_0 (water-filling)
- $P \leftarrow N_0 I + H_1 Q_1^{(t)} H_1^H$;
- $\Gamma \leftarrow H_0^H P^{-1} H_0 = V \Lambda V^H$;
- // Water-filling: allocate power p_i on eigenmodes of Γ
- $p_i \leftarrow \max(\mu - 1/\lambda_i, 0)$ s.t. $\sum_i p_i = P_0$;
- $Q_0^{(t+1)} \leftarrow V\,\mathrm{diag}(p_i)\,V^H$;
- // (B) Mirror descent of Q_1 (entropy geometry)
- $A \leftarrow P + H_0 Q_0^{(t+1)} H_0^H$, $G_1 \leftarrow H_1^H (A^{-1} - P^{-1}) H_1$;
- $Y \leftarrow \exp\left(\log Q_1^{(t)} - \eta G_1\right)$;
- $Q_1^{(t+1)} \leftarrow \begin{cases} Y, & \mathrm{tr}(Y) \le P_1, \\ \frac{P_1}{\mathrm{tr}(Y)} Y, & \text{otherwise.} \end{cases}$;
- // (C) Stop if changes are small
- **if** $\|Q_0^{(t+1)} - Q_0^{(t)}\|_F + \|Q_1^{(t+1)} - Q_1^{(t)}\|_F < \varepsilon$ **then**
  - **break**

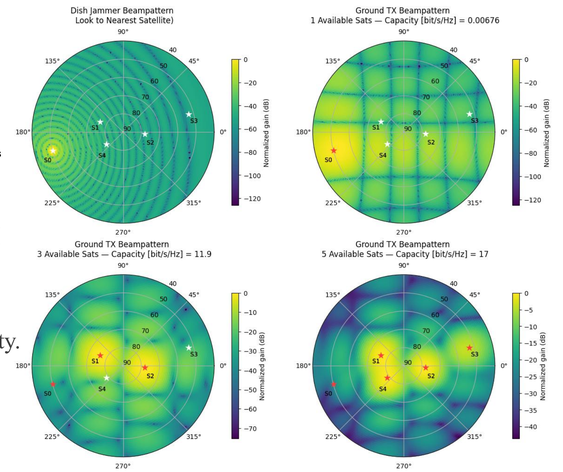
Output: $Q_0^\star$, $Q_1^\star$ and $J(Q_0^\star, Q_1^\star)$



Parameters	Values	Parameters	Values
Fc and B	10 GHz and 100 MHz	TX power	50 dBm
Jammer	6 × 6 URA, look to Z-up; Dish, look to nearest satllite	Jammer power	70 dBm
Desired TX	6 × 6 URA, look to Z-up	Elev, Az_degree	(30, 90), (0, 360)
Satellite antenna	6 × 6 URA, look to Earth's center	Atmospheric attenuation	3 dB

Results – Nullforming to Dish and Intelligent Jammer

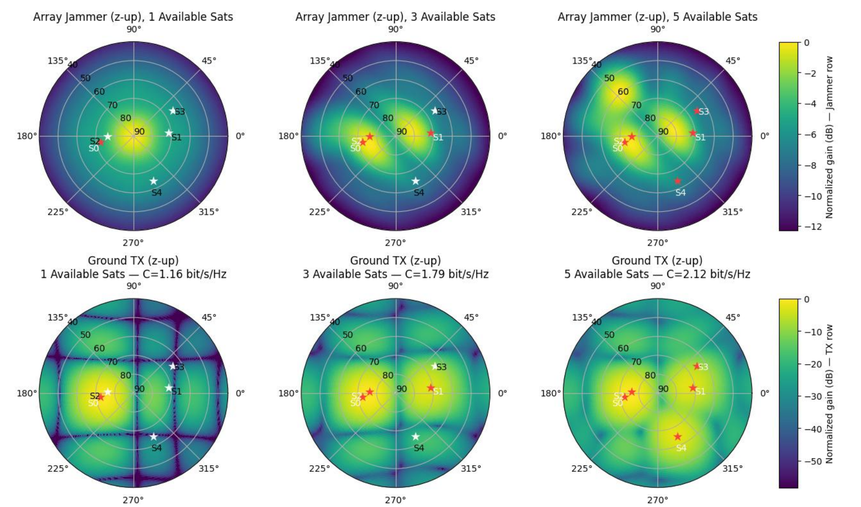


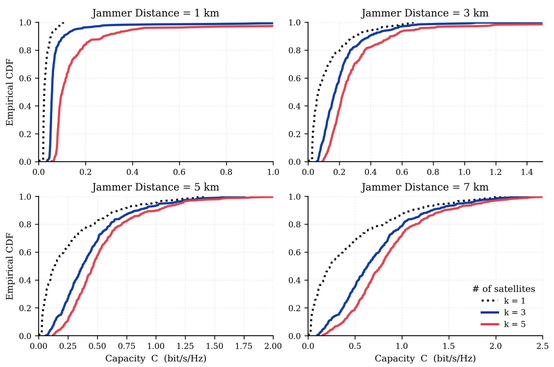


❑ **Dish jammer**

❑ Without beamforming capability.

❑ TX and Jammer distant 1 km





❑ **Array Jammer,**

❑ With beamforming capability.

❑ Jammer can learn the TX strategy.

❑ Jammer moves farther (1→7 km), its impact weakens.

❑ More satellites provide more strategy space and spatial diversity.